

Laboratory Work 6 Activity

Detecting Anomalies in
Time-Series Data Using Power BI

Part I. Data Preparation

Part 1: Data Preparation

1. Download the dataset and open Power BI Desktop.
2. Click Get Data → Text/CSV and import the dataset.
3. Ensure the following fields are correctly assigned:
 - Period → Date
 - Accommodation Type → Text
 - Travel → Text
 - Tourists → Whole Number

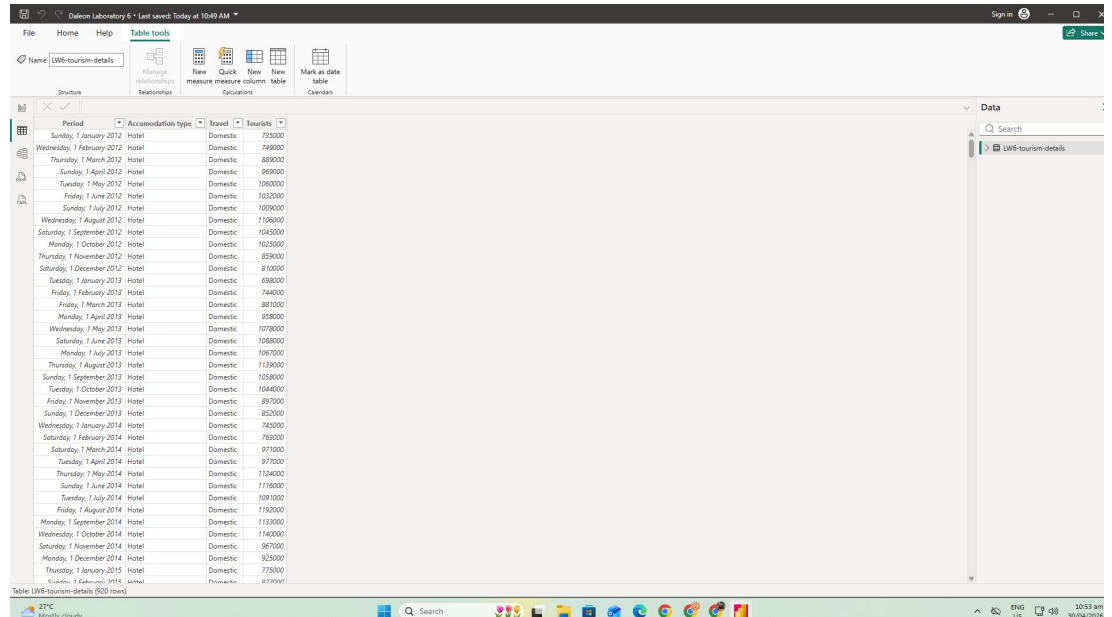


Table: LW6-tourism-details (500 rows)

Period	Accommodation type	Travel	Tourists
Sunday, 1 January 2012	Hotel	Domestic	723000
Wednesday, 1 February 2012	Hotel	Domestic	748000
Thursday, 1 March 2012	Hotel	Domestic	889000
Sunday, 1 April 2012	Hotel	Domestic	969000
Tuesday, 1 May 2012	Hotel	Domestic	1060000
Friday, 1 June 2012	Hotel	Domestic	1032000
Sunday, 1 July 2012	Hotel	Domestic	1009000
Wednesday, 1 August 2012	Hotel	Domestic	1106000
Saturday, 1 September 2012	Hotel	Domestic	1045000
Monday, 1 October 2012	Hotel	Domestic	1025000
Thursday, 1 November 2012	Hotel	Domestic	859000
Saturday, 1 December 2012	Hotel	Domestic	810000
Tuesday, 1 January 2013	Hotel	Domestic	698000
Friday, 1 February 2013	Hotel	Domestic	744000
Friday, 1 March 2013	Hotel	Domestic	881000
Monday, 1 April 2013	Hotel	Domestic	958000
Wednesday, 1 May 2013	Hotel	Domestic	1078000
Saturday, 1 June 2013	Hotel	Domestic	1088000
Monday, 1 July 2013	Hotel	Domestic	1067000
Thursday, 1 August 2013	Hotel	Domestic	1139000
Sunday, 1 September 2013	Hotel	Domestic	1058000
Tuesday, 1 October 2013	Hotel	Domestic	1044000
Friday, 1 November 2013	Hotel	Domestic	897000
Sunday, 1 December 2013	Hotel	Domestic	853000
Wednesday, 1 January 2014	Hotel	Domestic	745000
Saturday, 1 February 2014	Hotel	Domestic	763000
Saturday, 1 March 2014	Hotel	Domestic	977000
Tuesday, 1 April 2014	Hotel	Domestic	977000
Thursday, 1 May 2014	Hotel	Domestic	1124000
Sunday, 1 June 2014	Hotel	Domestic	1118000
Tuesday, 1 July 2014	Hotel	Domestic	1091000
Friday, 1 August 2014	Hotel	Domestic	1192000
Monday, 1 September 2014	Hotel	Domestic	1133000
Wednesday, 1 October 2014	Hotel	Domestic	1140000
Saturday, 1 November 2014	Hotel	Domestic	967000
Monday, 1 December 2014	Hotel	Domestic	825000
Thursday, 1 January 2015	Hotel	Domestic	775000

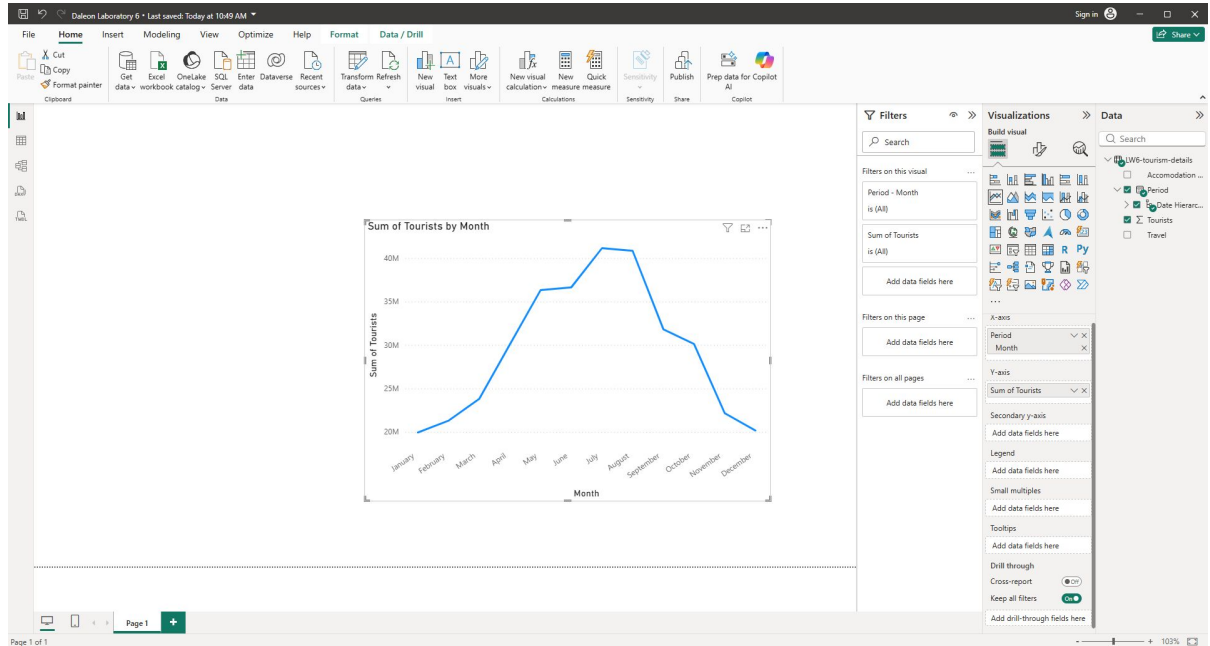
Part 2: Data Visualization

4. Insert a Line Chart visualization.

5. Drag:

- Period → Axis
- Tourists → Values

6. Expand the date hierarchy to show monthly data.



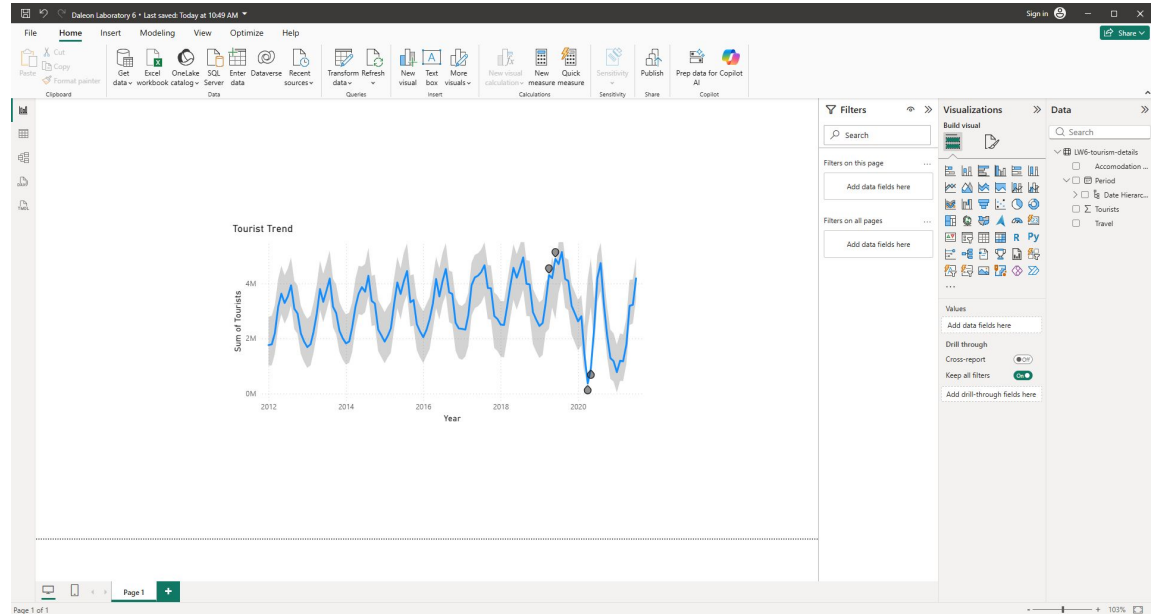
Part 3: Enable Anomaly Detection

7. Go to the Analytics Pane.

8. Click Find Anomalies → Add.

9. Observe:

- Highlighted anomaly points
- Expected value range (shaded region)



Part 4: Explain Anomalies

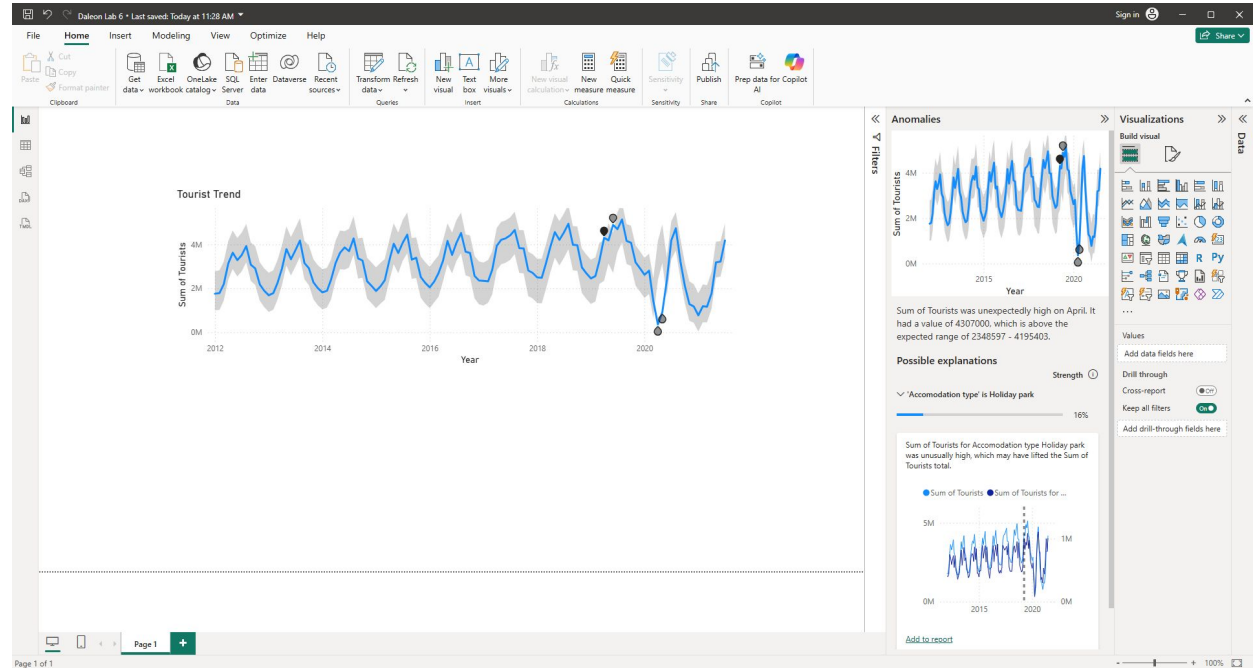
10. Drag the following into Explain By:

- Accommodation Type
- Travel

11. Click on any detected anomaly.

12. Analyze the Anomalies Pane:

- Identify whether the anomaly is high or low
- Review suggested explanations



Part 5: Interpretation and Analysis

1. What trend and seasonality patterns are observed?

From what I can see in the chart, the number of tourists follows a repeating pattern every year where it goes up and down. The higher values usually happen in the middle or later part of the year, while the lower values are mostly at the beginning. There's also a general increase from 2012 up to around 2019. But around 2020, there's a big drop that breaks the pattern, and after that, the data starts going up again.

2. Which anomalies are obvious vs. subtle?

The most obvious anomaly is the big drop around 2020 because it's way below the expected range and really stands out. There's also a noticeable spike after that, which is higher than expected. The subtle anomalies are the smaller points that are just slightly outside the shaded area. They're not as easy to notice unless you look closely.

3. What factors contributed to anomalies?

Based on the explanation panel, accommodation type like "Holiday park" seems to be one of the reasons for higher values. I think travel type also plays a role, since domestic and international travel can affect the number of tourists differently. These factors can change how many people are recorded in certain months.

4. Are the explanations logically valid? Why or why not?

Some of the explanations make sense because things like accommodation type can really affect tourist numbers. For example, if more people choose holiday parks, the total number can increase. But at the same time, I don't think the explanations are complete. The big drop around 2020 is probably caused by something bigger that the system didn't consider, so the explanation is only partially correct.

Part 6: Enhancement Task

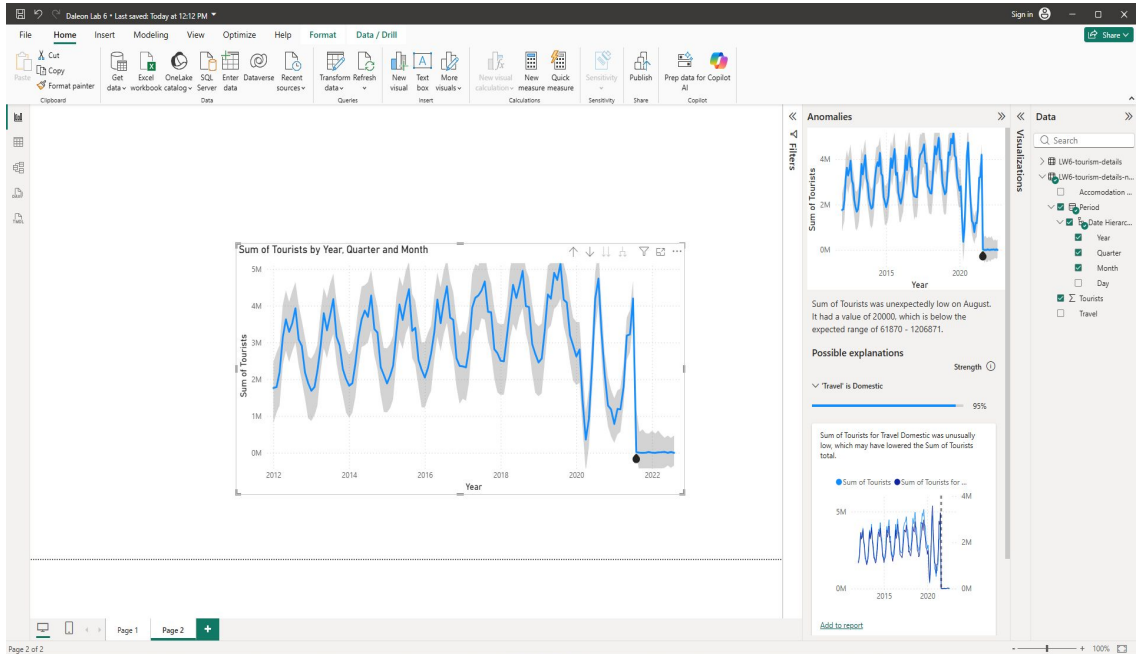
14. Add your own new anomaly dataset

(sample below) into your existing dataset.

15. Refresh the report and observe changes.

16. Compare:

- Before vs. after anomaly detection
- Newly detected anomalies



Part 6: Enhancement Task

Analysis:

The original dataset shows a clear seasonal pattern where tourist numbers go up and down in a consistent way every year. There is also a steady increase from 2012 to 2019, which shows that tourism was growing over time. One very noticeable anomaly appears around 2020, where the number of tourists suddenly drops far below the expected range. In the second visualization, after adding the 20 extra data entries, the pattern becomes less stable because of the extreme values that were introduced. There are more unusual spikes and drops, especially in 2021 and 2022, making the anomalies more obvious compared to the first result. Some of these changes are much more extreme, which affects how the data looks overall. The explanation pane points out factors like travel type and accommodation type as possible reasons, which makes sense based on the added data. However, these explanations are not always fully reliable since they are only based on patterns in the data, so it's still important to analyze the results carefully.